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September 29, 2026

Editors

Nano Letters

American Chemical Society

Dear Editors:

Please consider the enclosed Letter for publication in *Nano Letters*: “Dynamically Heterogeneous Mesoscale Arrest in a Protein Polymer with Reversible β -Sheet-like Order.”

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Significance. Dynamic arrest in a soft material is usually pictured as everything slowing down together. In the sequence-defined elastin-like polypeptide (VPAVG)₃₀ it does not work that way. Small-angle X-ray photon correlation spectroscopy (SA-XPCS) at 30 °C resolves a two-step correlation function whose fast mode is diffusive: its relaxation time scales as Q^{-2} , so it is described by a single effective diffusion coefficient. Over roughly 47 min that coefficient changes by less than a factor of two, while the fast-decay amplitude falls from 0.38–0.70 to 0.00–0.13 across 76–167 nm. What survives in the mobile population is therefore ordinary diffusive material, not partly arrested material: regions are either still fluid or already locked, and arrest advances by converting them rather than by stiffening them. It reaches the largest length scales first, with the fast fraction rising monotonically with Q at every elapsed time by nine standard deviations or more. Temperature-dependent SAXS/WAXS on the same construct resolves a reversible transition to a hierarchical assembly carrying β -sheet-like peaks at 1.38 and 0.72 Å⁻¹ that vanish on cooling, with seven consecutive 6–34 °C cycles recovering the low-temperature profile to within 9%.

Novelty. Prior work on gel-forming ELPs established arrested phase separation and network morphology from time-averaged scattering and rheology. The dynamical signature reported here, a diffusive fast mode whose time scale plateaus while its amplitude declines, has also been observed in a thermoreversible nanoparticle gel [D. Bahadur *et al.*, *J. Chem. Phys.* **151**, 104902 (2019)], and the manuscript cites that agreement rather than claiming the phenomenology as new. What is new is the length-scale decomposition, and what makes it quantitative is fitting the fast mode as a diffusion process rather than as a free relaxation time. Resolving the fast fraction separately in each Q bin shows that arrest is ordered by length scale, reaching 167 nm before 76 nm at every elapsed time, and follows the transfer to completion. The stretched correlation functions are shown to be diffusive rather than subdiffusive: the product of the stretching exponent with the Q exponent is 1.2, not the value of 2 that subdiffusive motion of identical scatterers requires. That decomposition needs microsecond framing: with the fast relaxation between 0.1 and 1.6 ms, a kilohertz acquisition returns only the slow component. We are equally explicit about the limits. The structural and dynamical observations come from separate experiments, so no causal link is claimed. The β -sheet assignment rests on WAXS peak positions rather than spectroscopy. The 1.6 s maximum delay bounds a fraction arrested on that scale, not an infinite-time nonergodicity factor. The isothermal series comes from one loaded sample.

Impact and urgency. The work spans sequence-defined nanomaterials, protein self-assembly, nonequilibrium soft matter, and coherent X-ray characterization, and gives a nanoscale structure–dynamics framework for responsive protein materials whose β -sheet-like order switches thermally without permanent aggregation. Because the behavior is sequence-encoded, the strategy transfers to other programmable protein polymers. The result rests on a capability that is not widely available. The XSPA-500k at 8-ID-I frames at 50 kHz, among the fastest in routine operation at a storage ring. The fast relaxation reported here, 0.1 to 1.6 ms, would fall at or below the shortest delay of a conventional kilohertz acquisition, and the amplitude transfer that is the paper’s central result would not be visible at all. High-speed detectors and upgraded storage rings have only recently made this practical on weakly scattering biological samples,

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so a concise, carefully bounded demonstration is timely. The reduced data and all analysis and figure-generation code are already public on GitHub (qzhang234/PubCode-2021-IDPP).

The manuscript is original, is not under consideration elsewhere, and has been approved by all authors. The authors declare no competing financial interest; AI-tool use is disclosed in the Acknowledgments. Thank you for your consideration.

Sincerely,

Qingteng Zhang, Ph.D., on behalf of all authors

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Suggested Reviewers

The following researchers have relevant expertise in elastin-like polypeptides, β -sheet-forming protein and peptide assemblies, small-angle X-ray scattering, and X-ray photon correlation spectroscopy.

Bradley D. Olsen

Alexander and I. Michael Kasser (1960) Professor of Chemical Engineering, Massachusetts Institute of Technology
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Expertise: elastin-like polypeptide gels, associative protein polymers, arrested phase separation, and neutron spin-echo measurement of relaxation in protein hydrogels.

Darrin J. Pochan

Professor of Materials Science and Engineering, University of Delaware
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Expertise: peptide and protein self-assembly, β -sheet hydrogel networks, and small-angle scattering and rheology of soft biomaterials.

Andrei Fluerasu

Lead Beamline Scientist, Coherent Hard X-ray Scattering, National Synchrotron Light Source II, Brookhaven National Laboratory
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Expertise: X-ray photon correlation spectroscopy and the microscopic dynamics of gels and arrested soft matter.

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Expertise: small-angle X-ray scattering and X-ray photon correlation spectroscopy of colloids and soft matter.